The documentation and process conversion measures necessary to comply with this revision shall be completed by 29 December 1994

INCH-POUND

MIL-S-19500/512B
29 September 1994
SUPERSEDING
MIL-S-19500/512A
8 June 1987

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON
SWITCHING TYPES 2N4029, 2N4033
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon transistors designed for use in high speed switching and driver applications. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figures 1 and 2 (TO-18 and TO-39).

1.3 Maximum ratings.

P_T T_A = +25°C		V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{OP} and T_{STG}
2N4029	2N4033					
W	W	V dc	V dc	V dc	A dc	°C
0.5 1/	0.8 2/	80	80	5	1.0	-55 to +200

1/ Derate at 2.86 mW/°C above T_A = 25°C.
2/ Derate at 4.56 mW/°C above T_A = 25°C.

1.4 Primary electrical characteristics at T_A = 25°C.

	h_{FE1}	h_{FE2}	h_{FE3}	h_{FE4}
LIMITS	V_{CE} = 5.0 V dc I_C = 100 µA dc	V_{CE} = 5.0 V dc I_C = 100 mA dc	V_{CE} = 5.0 V dc I_C = 500 mA dc	V_{CE} = 5.0 V dc I_C = 1.0 mA dc
MIN	50	100	70	25
MAX		300		

	$\|h_{fe}\|$	$V_{CE(sat)2}$	C_{obo}				
LIMITS	V_{CE} = 10 V dc I_C = 50 mA dc f = 100 MHz	I_C = 500 mA dc I_B = 50 mA dc	V_{CB} = 10 V dc I_E = 0 100 kHz < f < 1 MHz	t_d	t_r	t_s	t_f
		V dc	pF	ns	ns	ns	ns
MIN	1.5						
MAX	6.0	0.5	20	15	25	175	35

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

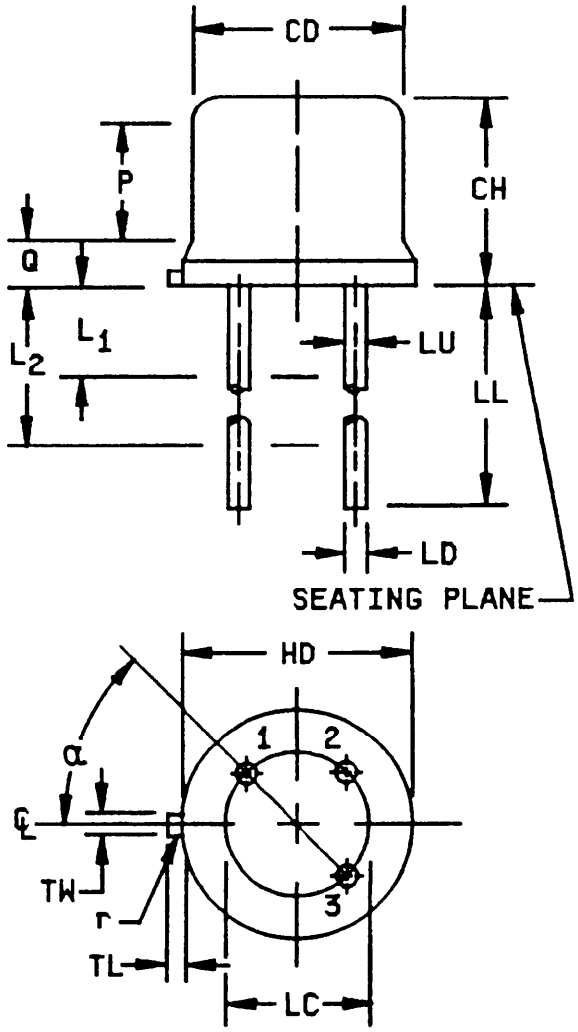
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Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	4
CH	.170	.210	4.43	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		5
LD	.016	.021	0.41	0.53	6
LL	.500	.750	12.70	19.05	
LU	.016	.019	0.41	0.48	6
L_1		.050		1.27	6
L_2	.250		6.35		6
TL	.028	.048	0.71	1.22	
TW	.036	.046	0.91	1.17	
P	.100		2.54		
Q		.040		1.02	
r		.010		0.18	10
α	45° TP		45° TP		

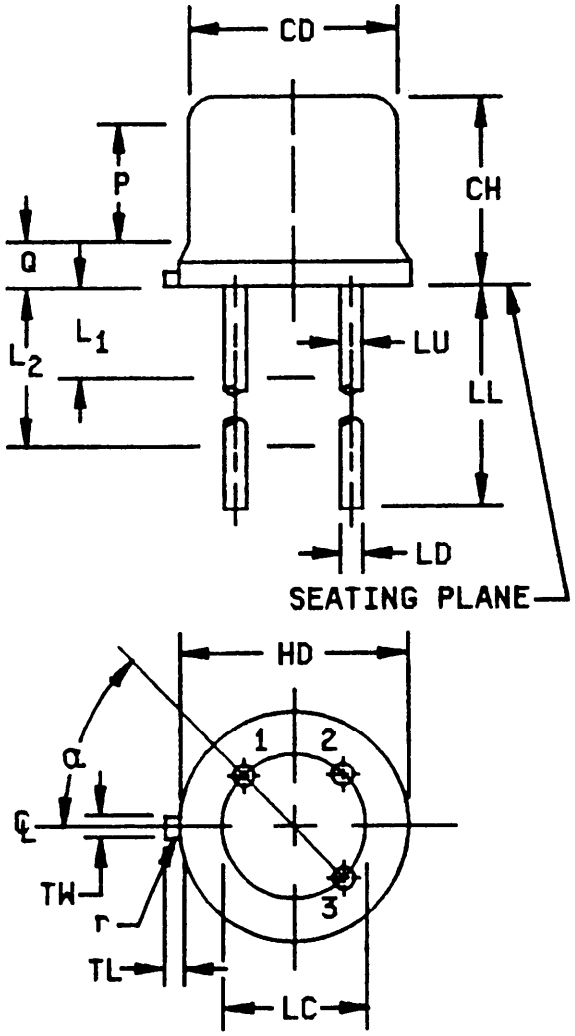


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Symbol CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
5. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
6. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Lead diameter shall not exceed .042 inch (1.07 mm) within L_1 and beyond LL minimum.
7. Lead 1 - emitter, lead 2 - base, lead 3 - collector.
8. Lead number three is electrically connected to case.
9. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Symbol r applied to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (type 2N4029).

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Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	4
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		5
LD	.016	.021	0.41	0.53	6
LL	.500	.750	12.70	19.05	
LU	.016	.019	0.41	0.48	6
L_1		.050		1.27	6
L_2	.250		6.35		6
TL	.029	.045	0.74	1.14	
TW	.028	.034	0.71	0.86	
P	.100		2.54		
Q		.050		1.27	
r		.010		0.18	10
α	45° TP		45° TP		



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Symbol CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
5. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
6. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Lead diameter shall not exceed .042 inch (1.07 mm) within L_1 and beyond LL minimum.
7. Lead 1 - emitter, lead 2 - base, lead 3 - collector.
8. Lead number three is electrically connected to case.
9. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Symbol r applied to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions (type 2N4033).

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitaion (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5M - Dimensioning and Tolerancing. (Dod adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in figure 1, (TO-18) for 2N4029 and figure 2, (TO-39) for 2N4033.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-S-19500, and as specified herein.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	h_{FE2} and I_{CBO1}	N/A
11	I_{CBO1}, h_{FE2}; ΔI_{CBO1} = 100% initial value or 2 nA dc, whichever is greater; Δh_{FE2} = 15% change from initial value.	I_{CBO1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 & 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 2 nA dc whichever is greater; Δh_{FE2} = 15% of initial value.	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 2 nA dc, whichever is greater; Δh_{FE2} = 15% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in MIL-S-19500;

JANS - - - V_{CB} = 10 V dc; JANTX, JANTXV - - - V_{CB} = 40 V dc;
2N4029 - - - P_T = 0.5 W
2N4033 - - - P_T = 0.8 W

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (endpoints) shall be in accordance group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 40 V dc; 2N4029 - P_T = 0.5 W; 2N4033 - P_T = 0.8 W; T_A = +25°C ± 3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 40 V dc; T_A = +125°C ± 25°C for 96 hours, P_T = 0.5 W for 2N4029, 0.8 W for 2N4033, or adjusted as required by the chosen T_A to give an average lot T_J = +275°C.
B5	2037	Test condition A (Au die interconnects only).
B6	3131	See 4.5.2.

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4.4.2.1 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	T_A = +30°C ± 5°C; V_{CB} = 40 V dc; P_T = 0.5 W for 2N4029, 0.8 W for 2N4033; No heat sink or forced-air cooling on the devices shall be permitted.
B5	3131	See 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (endpoints) shall be in accordance group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	T_A = +30°C ± 5°C; V_{CB} = 40 V dc; P_T = 0.5 W for 2N4029, 0.8 W for 2N4033; No heat sink or forced-air cooling on the devices shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance (to be performed for qualification inspection only). Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be .15 A dc.

b. Collector to emitter voltage shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C $\leq T_R \leq$ 35°C and recorded before the test is started.

e. Mounting arrangements shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 17.5°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage collector - emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Breakdown voltage collector - base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	80		V dc
Breakdown voltage emitter - base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	5		V dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		10	nA dc
Collector - emitter cutoff current	3041	Bias condition A; V_{CE} = 60 V dc; V_{BE} = 2.0 V dc	I_{CEX1}		25	nA dc
Base - emitter cutoff current	3061	Bias condition D; V_{BE} = 3.0 V dc	I_{EBO}		25	nA dc
Forward - current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 100 µA dc	h_{FE1}	50		
Forward - current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 100 mA dc; pulsed (see 4.5.1)	h_{FE2}	100	300	
Forward - current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE3}	70		
Forward - current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE4}	25		
Collector - emitter saturated voltage	3071	I_C = 500 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.15	V dc
Collector - emitter saturated voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.50	V dc
Collector - emitter saturated voltage	3071	I_C = 1.0 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		1.0	V dc

See footnote at end of table.

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TABLE I. Group A inspection - continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Cont.						
Base - emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$		0.9	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.2	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector - base cutoff current	3041	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		25	µA dc
Low - temperature operation		T_A = -55°C				
Forward - current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE5}	30		
Subgroup 4						
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 100 MHz	$\|h_{fe}\|$	1.5	6.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		20	pF
Input capacitance (output open - circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		80	pF
Pulse response:						
Delay time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see fig 3.)	t_d		15	ns
Rise time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see fig. 3)	t_r		25	ns

See footnote at end of table.

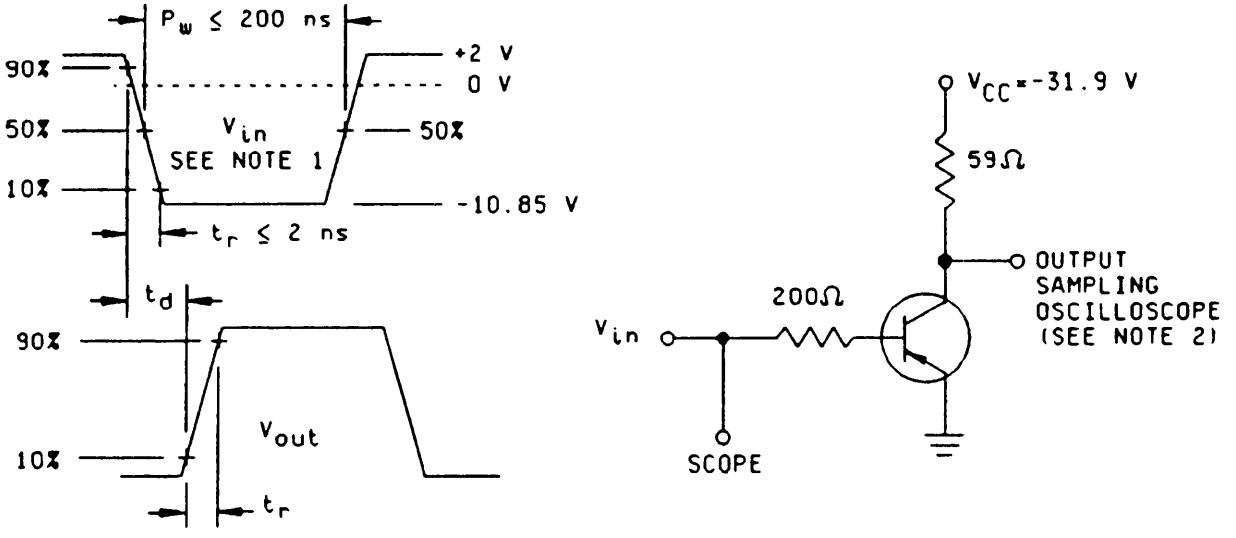
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TABLE I. Group A inspection - continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - cont.						
Storage time	3251	Test condition A; I_C = 500 mA dc; $I_{B1} = I_{B2}$ = 50 mA dc; (see fig. 4)	t_s		175	ns
Fall Time	3251	Test condition A; I_C = 500 mA dc; $I_{B1} = I_{B2}$ = 50 mA dc; (see fig. 4)	t_f		35	ns
Subgroups 5, 6 and 7						
Not applicable						

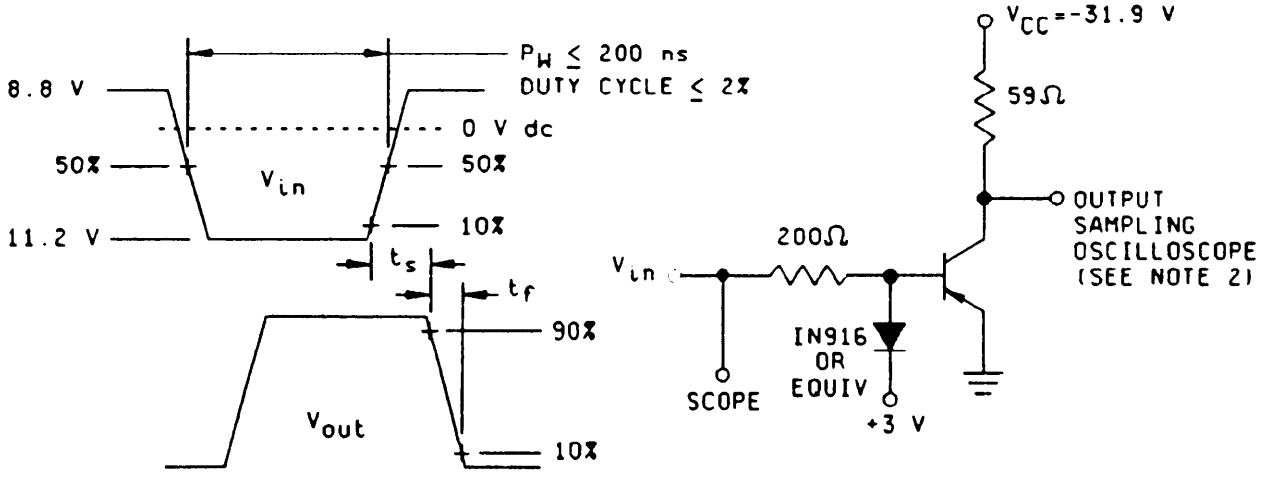
For sampling plan see MIL-S-19500.

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NOTES:
1. The rise time (t_r) or applied pulse shall be ≤ 2 ns, duty cycle ≤2% and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 3. Test circuit for measuring t_d and t_r times.



NOTES:
1. The rise time (t_r) or applied pulse shall be ≤ 2 ns, duty cycle ≤2% and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 4. Test circuit for measuring t_s and t_f times.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type designation and quality product level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/512B	2. DOCUMENT DATE (YYMMDD) 94/09/29

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON SWITCHING TYPES 2N4029, 2N4033 JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Joe Barrows	b. TELEPHONE (Include Area Code) (1) Commercial 301-731-8694	(2) AUTOVON
c. ADDRESS (Include Zip Code) NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center Code 310.A Greenbelt, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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